

NYC Yellow Taxi Data Analysis

Big Data Analytics

Group 55

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[GitHub Project](#)

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1 Introduction & Background

1.1 Problem Statement and Research Motivation

In an increasingly data-driven world, it is essential to respond effectively to customer expectations. Big Data Analytics (BDA) provides powerful tools to analyze vast amounts of transportation data to improve service delivery and operational efficiency. The problem addressed in this project is multifaceted: 1) clustering similar pickup and drop-off areas to uncover spatial patterns in taxi usage; 2) forecasting the number of trips per hour to better anticipate demand and optimize taxi availability; 3) predicting trip prices to provide customers with reliable cost estimates before their journey; 4) graph analysis to explore the structural relationships within the taxi trip network and identify key transit routes.

These components together aim to enhance customer satisfaction and support data-informed decision-making for taxi service providers and city planners. To effectively tackle these challenges, we make use of Apache Spark's distributed processing capabilities, combining Spark SQL, MLlib and GraphFrames to efficiently handle large-scale data and implement a diverse range of analytical techniques.

We selected the NYC Taxi trip dataset due to its richness, scale, and versatility. Its diversity enables us to explore a variety of analytics methods - from clustering and time series forecasting to predictive modeling and graph-based analysis - highlighting the broad applicability and value of BDA in addressing real-world transportation challenges.

This topic combines societal impact by addressing urban mobility issues, business value by supporting demand-driven pricing and resource allocation, and technical depth through the handling of large-scale, heterogeneous time-series data using distributed computing tools like Spark.

2 Data Collection & Preprocessing

2.1 Data Sources and their Characteristics

The primary dataset used in this project is the New York City Taxi and Limousine Commission (TLC) Trip Record Data, which is publicly available through the NYC Open Data platform. This dataset contains detailed trip-level records collected from both yellow and green taxi services operating within New York City. For our analysis, we focus specifically on the yellow taxi trip data from 2022 to 2024.

In addition to the taxi trip data, weather data is incorporated to enhance forecasting accuracy by accounting for external factors that may influence taxi demand. The data was obtained from the National Centers for Environmental Information

The datasets are provided as parquet files and are structured, containing key attributes such as pickup and drop-off timestamps and locations, trip distances, fare amounts, and other relevant details necessary for analysis. The weather data is also a structured dataset.

2.2 Data Cleaning & Preprocessing Steps

During the exploratory analysis, several histograms, boxplots and heatmaps of correlations between variables were created, with the aim of understanding the distribution, relationships and identifying possible anomalies in the data. This process uncovered

multiple quality issues: some columns had missing values in fields that are essential for reliable analysis; numerical variables presented extreme outliers; and there were temporal inconsistencies, such as trips with pickup times after drop-off times. We also found other irregularities, such as negative values in columns that should only contain positive numbers, and discrepancies in categorical data.

Since we are dealing with a large-scale dataset, removing a small fraction of problematic rows would not affect the analysis. Therefore, we decided to eliminate null values, filter out the most extreme outliers, and remove records containing invalid timestamps or other inconsistent information. These steps were fundamental to ensure data quality and consistency, enabling robust and meaningful downstream analysis.

3 Methodology & Tools

3.1 Forecasting Techniques

To forecast the number of taxi trips per hour in New York City, we developed a forecasting model based on time, consequences, and calendar variables. The model was built to identify regular demand patterns and predict their changes over time, allowing for more efficient use of available resources in an urban environment.

We began by loading the cleaned dataset with Spark, which allowed us to work efficiently with large volumes of data. We also imported the necessary Python libraries to support our analysis and modeling. To help the model understand patterns in taxi usage, we created new time-based features from the pickup time, such as hour of the day, day of the week, and whether the trip happened on a weekend.

Next, we grouped the data by hour to get the total number of trips in each hour. We enriched this information by adding weather data—such as temperature, humidity, and precipitation—as well as a flag for public holidays, since these factors can affect taxi demand. To avoid problems with missing time periods, we created a full sequence of hourly timestamps and filled in any hours without trips by setting the count to zero. This ensured a complete and continuous time series.

To prepare the data for machine learning, we added lag features (previous values of taxi demand) and rolling averages. These help the model learn from past trends.

The data was sorted in chronological order and split into three distinct subsets: training, validation, and testing. The training set included all data from the years 2022 and 2023 and was used to fit the candidate models. The validation set, covering the first half of 2024, was used to compare the performance of different models and to fine-tune hyperparameters based on used metrics. Finally, the second half of 2024 was reserved as the test set, used exclusively for the final evaluation of the selected model, simulating forecasts on entirely unseen data. This approach respects the natural temporal order of events and avoids any data leakage between subsets.

We then trained three different models using Spark MLlib: Random Forest, Gradient Boosted Trees (GBT), and Linear Regression. We used cross-validation to find the best model based on prediction accuracy, measured by root mean square error (RMSE).

3.2 Data Mining Techniques

3.2.1 Clustering Analysis

To explore potential groupings within the data, we conducted a clustering analysis following a structured approach. We began by handling missing values and performing a simple feature analysis to understand data distributions and correlations. A correlation matrix was created to identify potential relationships among variables as in Table 4.

The data was then normalized using StandardScaler to ensure all features contributed equally to the clustering process. Due to computational limitations in Databricks, a representative sample of 50,000 records was extracted from the full dataset to proceed with clustering.

Initially, we applied K-means clustering with $k=3$, $k=4$, and $k=5$. Among these, $k=3$ provided slightly better results as in Figure 5, but the overall outcome was still unsatisfactory—clusters remained poorly defined and significantly overlapped. This limitation led us to adopt a density-based approach, DBSCAN, which proved to be more effective for this dataset, as it can handle clusters of varying shapes and densities and is less sensitive to outliers.

We tested DBSCAN using both PCA and t-SNE for dimensionality reduction. PCA results showed poor visual separation as we can see in Figure 6, while t-SNE revealed much clearer and more meaningful cluster separation, as illustrated in Figure 7, highlighting the effectiveness of combining DBSCAN with a non-linear projection technique.

3.3 Machine Learning Techniques

3.3.1 Regression Analysis Price Prediction

This section focuses on building predictive models that estimate trip prices based on various trip features.

To identify the most relevant features, we use mutual information scores for categorical variables and correlation analysis for numerical features, complemented by feature importance from tree-based models.

For modeling, we start with a simple linear regression as a baseline, then employ more powerful algorithms such as Random Forest (RF) and Gradient Boosted Trees (GBT) regressors to improve prediction accuracy. The dataset was split into training, validation, and test sets. The validation set was used to compare different models and feature sets, while the test set was reserved only for evaluating the final model’s performance.

In addition, we use Grid Search combined with cross-validation to fine-tune hyperparameters for RF and GBT models. Model training and evaluation are performed in a distributed environment using PySpark, ensuring scalability on large datasets.

MLflow is used to track experiments, parameters, and results systematically. Finally, residual plots help us visually assess model performance and identify areas for improvement.

3.3.2 Graph Analysis

We constructed a directed graph using pickup and drop-off location IDs, where each node represents a location and each directed edge represents trips between them.

The following methods were applied: 1) in-degree and out-degree centrality were computed to identify the most active pickup and drop-off zones in the city; 2) label Prop-

agation Algorithm (LPA) was used to detect communities or clusters of zones that are frequently connected by trips; 3) K-means clustering was applied to the locations (graph vertices) to uncover additional spatial groupings based on their trip-related characteristics.

These methods help to identify both structurally important zones and naturally emerging communities within the city’s taxi network.

4 Results

4.1 Forecasting Results

Table 1 presents the values of the three main evaluation metrics (MAE, RMSE, and R^2) for each of the models considered: Linear Regression (LR), Random Forest (RF), and Gradient Boosted Trees (GBT). The results show a clear superiority of the GBT model, which achieved the lowest mean absolute error ($MAE = 259.81$) and the lowest root mean squared error ($RMSE = 371.85$), as well as the highest R^2 value (0.9742), indicating a strong ability to explain the variability in hourly demand.

Despite the overall strong performance, some relevant limitations were identified (Figure 1). In particular, the model struggled to predict demand peaks above 8,000 trips per hour, showing a tendency to underestimate demand during high-intensity periods. Slight temporal shifts were also observed in the forecasts during abrupt transitions.

4.2 Regression Analysis Results

We started the regression analysis by performing feature selection. Any redundant variables or variables with low mutual information scores and negligible differences in average trip prices were removed. Figure 2 in Appendix shows the mutual information scores for the categorical features (except for pick up and drop off locations).

All numerical features, namely `trip_distance`, `extra`, and `tolls_amount`, were retained, as each showed at least some correlation with the target variable and were therefore kept at this stage. Figure 3 in Appendix shows scatter plots illustrating the relationships between these features and the target.

After an initial iteration of the Random Forest (RF) and Gradient Boosted Trees (GBT) models, we found that `passenger_count` and `is_weekend` had low feature importance across both models. These features were subsequently removed, and RF and GBT models were retrained using the updated feature set (Features 4). Feature sets are presented in Table 2 in Appendix.

We evaluated six models: two Linear Regression models, two Random Forest models, and two Gradient Boosted Trees (each using different sets of features). Table 3 in Appendix summarizes their performance metrics. Among them, the GBT model using the third feature set (Features 3) achieved the best overall performance with mean absolute error of 2.35 \$ and was selected as the final model.

To evaluate model performance, we calculated metrics on the test data and visualized the results using residual plot presented in Figure 4 in Appendix. The residual plot reveal the presence of some trips for which the model’s predictions are less accurate. However, there are no strong or systematic patterns in the residuals. For instance, we notice a gap around the predicted value of 150, but overall, the distribution appears balanced. This suggests that the final model performs consistently across the majority of data points.

4.3 Clustering Results

The final clustering outcome, obtained using DBSCAN and visualized through the t-SNE projection, revealed a meaningful and well-separated distribution of clusters. Unlike previous attempts with K-means, where cluster boundaries were often ambiguous, the density-based approach succeeded in identifying compact separated groups of records. As shown in Figure 7, the t-SNE transformation enhanced the cluster structure by projecting high-dimensional relationships into a 2D space, where distinct patterns and subgroupings became more visually apparent. This result confirmed that DBSCAN, combined with a non-linear dimensionality reduction technique like t-SNE, was the most appropriate choice for capturing the complexity of the dataset.

4.4 Graph Analysis

The graph-based analysis revealed some insights into the structure of taxi trip flows in New York City. Firstly, among the top 10 locations with the highest in-degrees and out-degrees, 7 locations overlapped, indicating that certain zones serve as both major pickup and drop-off hubs. These locations are highlighted in Figure 8 in Appendix.

The Label Propagation Algorithm (LPA) identified only one community, suggesting that the taxi trip network is highly interconnected, with no clearly separated clusters. However, only a subgraph with edges where the trip count is more than 1000 were analyzed.

K-means clustering of locations produced three distinct groups based on trip dynamics: 1) a majority group characterized by low in/out degrees but high average trip distances, likely representing residential zones with fewer but longer trips; 2) a small cluster with high degree values but low average distance and fare amounts, possibly corresponding to dense commercial or downtown areas with frequent, short trips; 3) a middle group with moderate degree values and relatively low trip distances and amounts, possibly reflecting mixed-use zones.

These results illustrate how graph-based and clustering techniques can expose underlying spatial and behavioral patterns in transportation networks.

5 Conclusion

One of the aspects we are most proud of from a creativity standpoint is how we combined multiple Big Data techniques, like forecasting, predictive modeling, clustering, and graph analysis, to explore the NYC Taxi dataset from different angles. We also learned how critical data preprocessing is when working with real-world datasets that are large and noisy.

One of the challenges was handling high-dimensional categorical data, especially pickup and drop-off location IDs. These features are important but difficult to incorporate directly in regression models due to cardinality. We addressed this by using ensemble models like Random Forest and GBT, which can handle such features better.

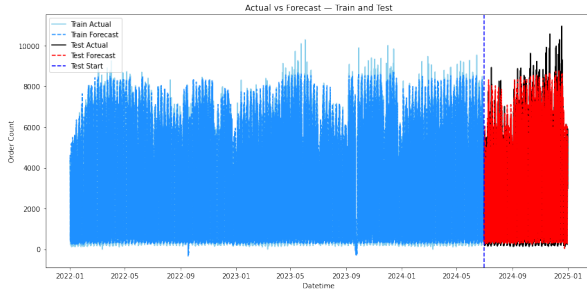
Another challenge emerged in graph community detection: the Label Propagation Algorithm identified only a single community, limiting its usefulness. A potential solution would be to try alternative community detection algorithms or modify the graph filtering.

A Appendix

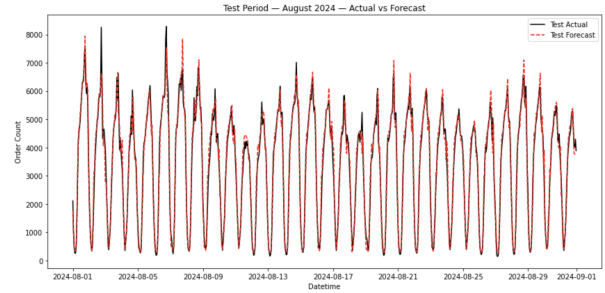
A.1 Forecasting

Model	MAE	RMSE	R ²
Gradient Boosted Trees	259.81	371.85	0.9742
Linear Regression	440.28	595.48	0.9337
Random Forest	359.15	493.79	0.9544

Table 1: Performance metrics summary for each predictive model.



(a) Full dataset: actual vs forecast (train and test).



(b) Zoom into test period — August 2024.

Figure 1: Forecast performance on train and test sets. The left plot shows the complete time range, while the right plot focuses on daily dynamics during a representative month (August).

A.2 Regression Analysis

Feature	MI Score
improvement_surcharge_indexed	0.455663
hour	0.271016
has_toll	0.207346
airport_fee	0.181585
congestion_surcharge_indexed	0.109734
is_weekend	0.037568
mta_tax_indexed	0.026799
day_of_week	0.025195
passenger_count	0.004899
is_shared_ride	0.004638
is_holiday	0.002888

Figure 2: Mutual information scores of categorical features.

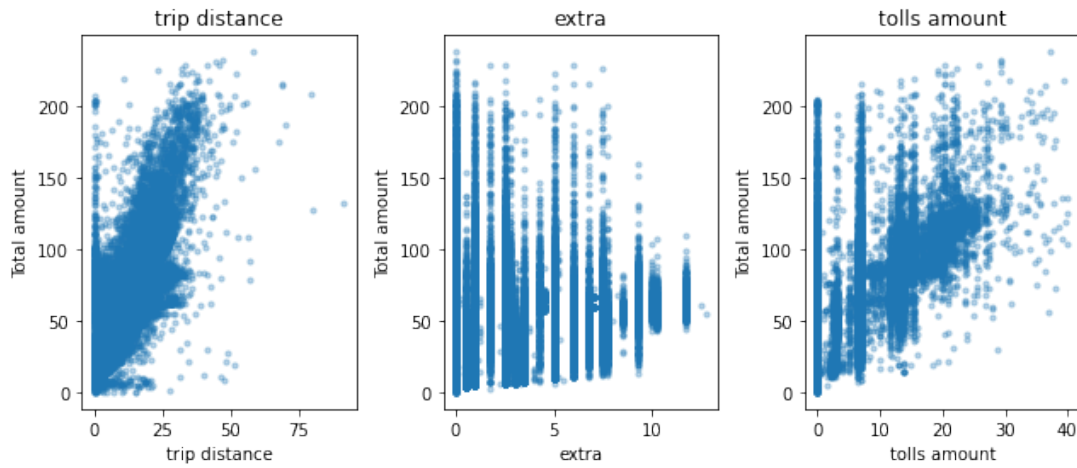


Figure 3: Scatter plots of numerical features against total amount

Feature	Description
Features 1	trip_distance
Features 2	trip_distance, extra, tolls_amount
Features 3	passenger_count, PULocationID, DOLocationID, airport_fee, is_weekend, hour, mta_tax_indexed, congestion_surcharge_indexed, trip_distance, extra, tolls_amount
Features 4	Features 3 except passenger_count and is_weekend

Table 2: Feature sets used for the regression models

Model	Parameters	Features	RMSE	R^2	MAE
Linear Regression	Default	Features 1	6.53	0.85	3.86
Linear Regression	Default	Features 2	5.95	0.88	3.60
Random Forest Regression	NumTrees 20, MaxDepth 5, MaxBins 32	Features 3	5.71	0.89	3.71
Random Forest Regression	NumTrees 50, MaxDepth 5, MaxBins 32	Features 4	6.02	0.88	4.05
GBT Regression	MaxDepth 5, MaxBins 32	Features 3	4.39	0.93	2.35
GBT Regression	MaxDepth 5, MaxBins 32	Features 4	4.97	0.92	2.9

Table 3: Model performance comparison

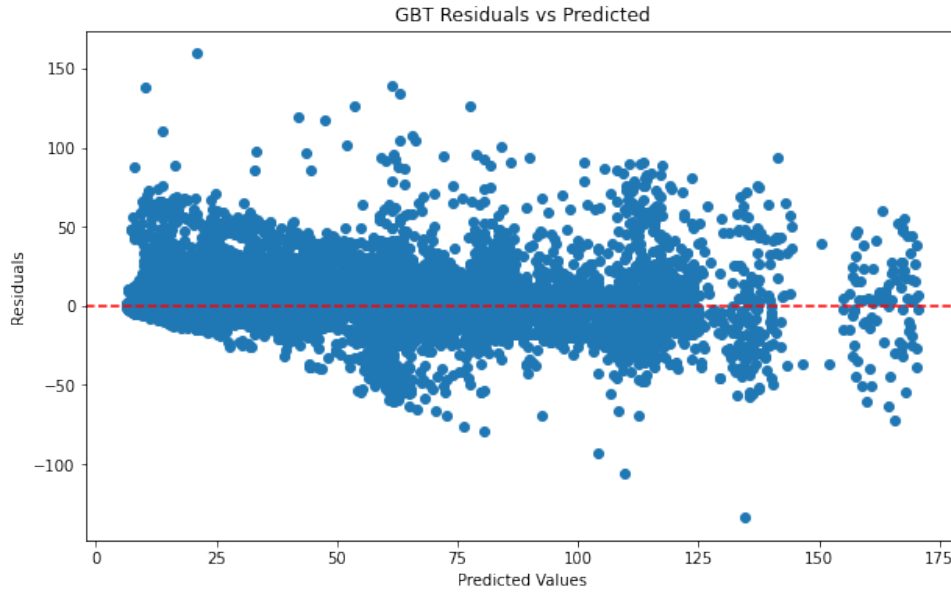


Figure 4: Residual plot of final regression model

A.3 Clustering

	trip_distance	fare_amount	extra	mta_tax	tip_amount	tolls_amount	improvement_surcharge	total_amount	congestion_surcharge	pickup_hour_decimal	trip_duration_minutes	trip_speed_mph
trip_distance	1.000000	0.943946	0.319051	-0.126552	0.592589	0.632882	0.093409	0.923529	-0.322321	-0.046407	0.239495	0.046940
fare_amount	0.943946	1.000000	0.364485	-0.183618	0.625218	0.618589	0.236414	0.977355	-0.318956	-0.025750	0.260134	0.027860
extra	0.319051	0.364485	1.000000	-0.017117	0.310273	0.402960	0.280441	0.434240	-0.098024	-0.019673	0.077417	0.006207
mta_tax	-0.126552	-0.183618	-0.017117	1.000000	-0.097836	-0.258550	-0.013361	-0.180106	0.169665	0.016054	-0.024139	-0.009667
tip_amount	0.592589	0.625218	0.310273	-0.097836	1.000000	0.480639	0.181250	0.744840	-0.124623	0.003446	0.154537	0.014595
tolls_amount	0.632882	0.618589	0.402960	-0.258550	0.480639	1.000000	0.100490	0.699841	-0.124783	-0.041930	0.141947	0.023054
improvement_surcharge	0.093409	0.236414	0.280441	-0.013361	0.181250	0.100490	1.000000	0.269526	-0.079673	0.009399	0.022306	-0.003700
total_amount	0.923529	0.977355	0.434240	-0.180106	0.744840	0.699841	0.269526	1.000000	-0.278472	-0.013634	0.249323	0.026714
congestion_surcharge	-0.322321	-0.318956	-0.098024	0.169665	-0.124623	-0.124783	-0.079673	-0.278472	1.000000	-0.000670	-0.049697	-0.025596
pickup_hour_decimal	-0.046407	-0.025750	-0.019673	0.016054	0.003446	-0.041930	0.009399	-0.013634	-0.000670	1.000000	-0.000152	-0.007292
trip_duration_minutes	0.239495	0.260134	0.077417	-0.024139	0.154537	0.141947	0.022306	0.249323	-0.049697	-0.000152	1.000000	-0.002445
trip_speed_mph	0.046940	0.027860	0.006207	-0.009667	0.014595	0.023054	-0.003700	0.026714	-0.025596	-0.007292	-0.002445	1.000000

Table 4: Correlation Matrix in Cluster Analysis

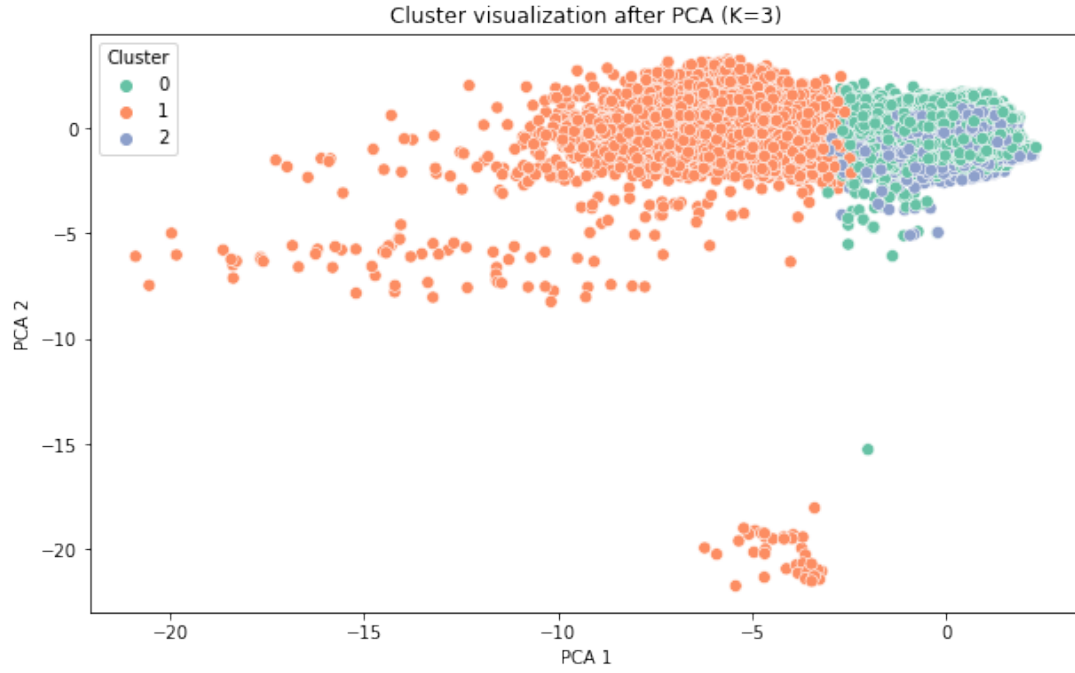


Figure 5: KMeans vizualization with PCA (K=3)

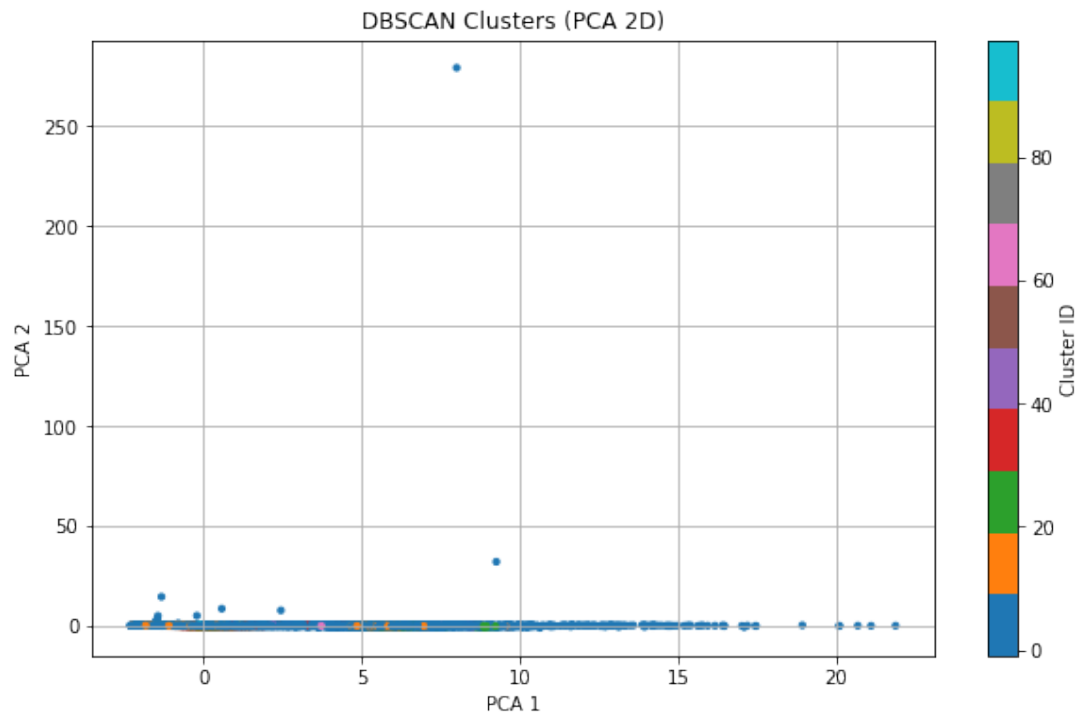


Figure 6: DBSCAN with PCA

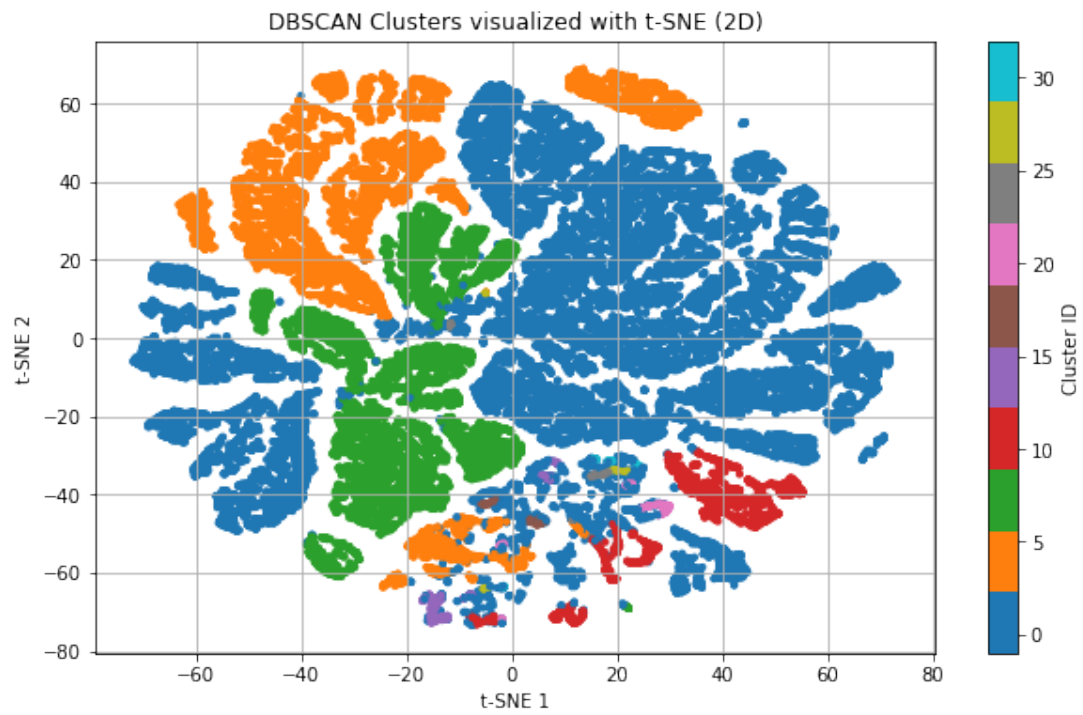


Figure 7: DBSCAN with T-SNE (Final Cluster Algorithm)

A.4 Graph Analysis

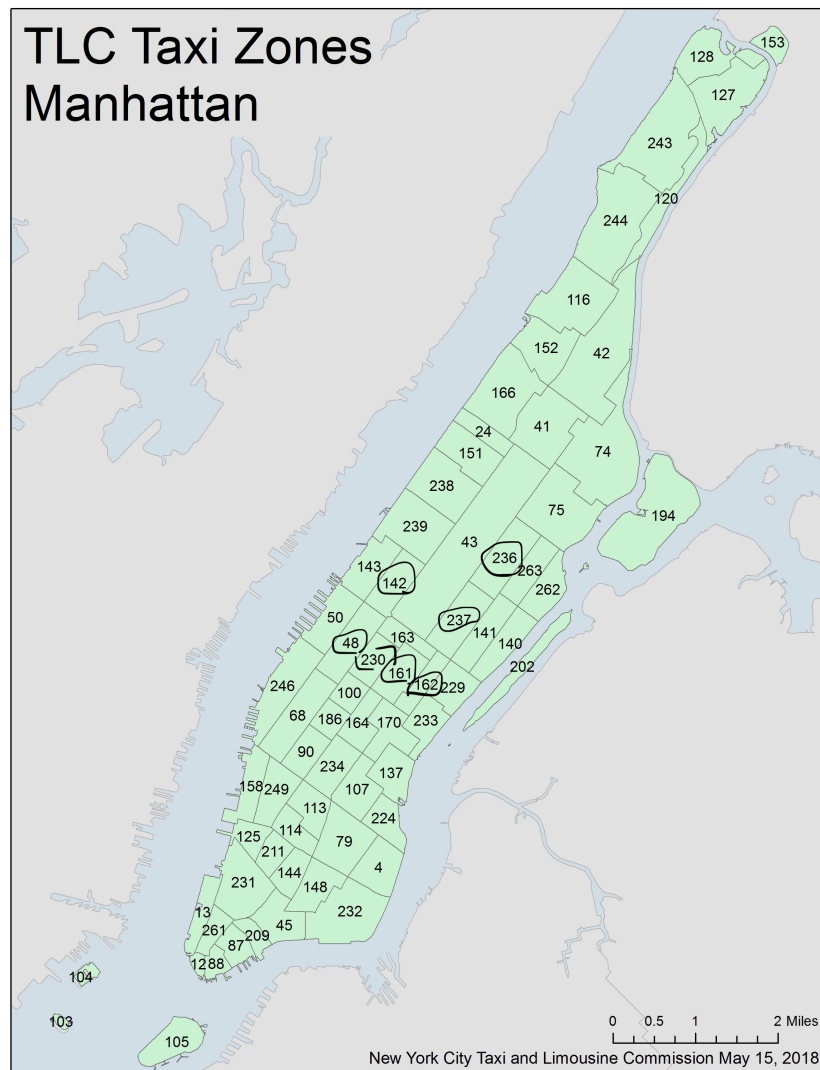


Figure 8: Taxi zones that appear in the top 10 for both in- and out-degree.